120 Days of Quant Finance

Week 6: Stochastic Processes & Dynamic Asset Modeling
Day 40: Dynamic Hedging and Delta Hedging Intuition

Arnav Garg

March 2026

Introduction

Options are nonlinear financial instruments whose risk profiles evolve with market conditions. Managing option risk requires continuously adjusting positions to offset changes in the underlying asset price. Dynamic hedging, and in particular delta hedging, provides a systematic framework for this task.

Delta Hedging

The Delta of an option is defined as:

$$\Delta = \frac{\partial V}{\partial S},$$

where V is the option value and S is the underlying asset price.

Delta hedging involves holding an offsetting position of $-\Delta$ units in the underlying asset to neutralize first-order exposure to price movements.

Why Hedging Must Be Dynamic

Option Delta is not constant and varies with the underlying price, time to maturity, and volatility. This sensitivity of Delta is measured by Gamma:

$$\Gamma = \frac{\partial^2 V}{\partial S^2}.$$

As a result, delta hedges must be continuously rebalanced, leading to the concept of dynamic hedging.

Dynamic Hedging and Risk Neutrality

Under idealized assumptions of continuous trading and frictionless markets, a dynamically delta-hedged option portfolio becomes locally riskless. Such a portfolio must earn the risk-free rate, providing the foundation for risk-neutral pricing.

Gamma, Theta, and Hedging Costs

Even when delta-hedged, an option portfolio remains exposed to Gamma and Theta. Positions with high Gamma require frequent rebalancing and typically exhibit negative time decay, while positions with low Gamma may generate steady income but carry tail risk.

Limitations of Dynamic Hedging

Dynamic hedging is imperfect in practice due to discrete rebalancing, transaction costs, liquidity constraints, and sudden market jumps. These limitations explain why hedging strategies can fail during periods of market stress.

Conclusion

Dynamic hedging transforms directional price risk into time decay and rebalancing costs. While perfect hedging is unattainable in real markets, Greek-based dynamic hedging remains a central tool in options risk management.